Pragyan Kadel

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Houston, TX

PROFESSIONAL SUMMARY

Data science graduate with experience in Python, SQL, ETL, machine learning, and large-scale healthcare data analysis. Strong background in predictive modeling, data quality, and transforming complex datasets into reliable analytical pipelines.

SKILLS

Programming Languages	Python, SQL, R, Bash
Data Quality & Analysis	Exploratory Data Analysis, Validation, Reconciliation, Profiling, Source-to-Target Validation, Root Cause Analysis
ETL & Data Engineering	ETL Pipelines, ETL Testing, Data Transformation Validation, Data Warehousing, Snowflake
Big Data & Orchestration	PySpark, Apache Airflow, Distributed Data Processing
Databases	PostgreSQL, MySQL, NoSQL
Machine Learning	ANNs (FNNs, CNNs, RNNs, LSTMs), SVMs, RVMs, Random Forest

PROFESSIONAL EXPERIENCE

Graduate Research Assistant – Machine Learning & Statistical Modeling May 2025 – Present
Department of Mathematics, Clarkson University Potsdam, New York

- Developed predictive pipelines for healthcare datasets exceeding 40Million+ records, integrating multi-source data (Electronic Health Records & Census 2020) to analyze cancer lethality and socio-demographic disparities.
- Conducted performance benchmarking across Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), Relevance Vector Machines (RVMs), and Random Forest to optimize diagnostic accuracy for prostate cancer screening.
- Used genetic biomarkers and longitudinal PSA data to extract features for early-stage prostate cancer grade prediction.
- Used classification models; namely SVMs and RVMs, hyper tuned them to optimize efficiency based on data spread and applied data balancing techniques such as SMOTE and Bootstrapping

Academic Python Trainee April 2022 – November 2022
Research & Development, DigiSchool Global Kathmandu, Nepal

- Delivered a Python programming course to high school students issued by NCC, UK that included data manipulation techniques using pandas and numpy, file handling and object-oriented programming (OOP).
- Developed GUI based ETL pipeline to handle high school students' data of over 1200 students and visualize accounts data using Tableau.
- Designed and implemented interactive python programming platform to facilitate teaching programming skills to middle school students in 7 schools in Kathmandu, Nepal.

PROJECTS

A Clinical Evaluation of Longitudinal PSA and Sociodemographic Features for Predicting Prostate Cancer Incidence

- Integrated 5 databases with 40M+ real-time records (PSA screening + socio-demographic) into Snowflake; engineered 5 clinical features and improved prostate cancer likelihood prediction by 27% after hyper-tuning. Collaborated with Henry Ford Health System.

SVM Models for Prostate Cancer Prediction Using PVT1 Biomarkers and Baseline PSA

- Integrated PSA, genetic (exon biomarker), and biopsy (Gleason) data of 144 patients; trained SVMs/RVMs, tuned hyperparameters, and improved prostate cancer classification accuracy by 20%+ vs. industry standard across racial groups.

EDUCATION

MS in Applied Data Science Aug 2024 – May 2026
Clarkson University, Potsdam, New York *GPA: 4.0*
Coursework: Data Mining, Applied ML, Big Data Architecture, NLP, Data Warehousing, DB Modeling, Probability & Statistics, Information Visualization

BSc in Ethical Hacking and Cybersecurity Sep 2019 – Aug 2022
Coventry University (Softwarica College of IT & E-Commerce), Kathmandu, Nepal *72.75%*
Coursework: Algorithms, Operating Systems, Security & Networks, Networked System Architectures, Systems Security

PRESENTATIONS

Comparing ML Models for Prostate Cancer Prediction Using PVT1 Biomarkers and Baseline PSA Apr 21, 2026
American Association of Cancer Research (AACR) Annual Meeting, San Diego, California

Prostate Cancer Prediction Using PVT1 Biomarkers and Baseline PSA Jul 31, 2025
Research and Project Showcase (RAPS), Clarkson University, Potsdam, NY

RF Model Integrating PSA & Sociodemographic Features to Predict Likelihood and Lethality of Prostate Cancer Jan 13, 2026
LEAP Summit, Clarkson University, Potsdam, NY

CERTIFICATIONS

Alteryx Data Analytics Foundational Certification April 2025
AWS Certified Solutions Architect – Associate (SAA-C03) May 2026